Lab 1: Requirements Engineering & UML Use-Case Modelling

Pharmacy Medicine Pickup System

Complete Solution

1. Requirements Table

The assignment requires exactly five Functional Requirements and two Nonfunctional Requirements, with measurable acceptance criteria and rationales.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow customers to search for and select required medicines.	High	For a valid medicine search, matching medicines are displayed and selected medicines appear in the order summary.	Core functionality needed to create a medicine order.
FR-002	Functional	The system shall allow customers to upload or scan a prescription for prescription-only medicines when required.	High	A valid prescription file/image can be attached to the relevant order before submission.	Supports prescription verification for regulated medicines.
FR-003	Functional	The system shall allow customers to place a medicine pickup order containing the selected medicines and required prescription.	High	After valid order submission, the system generates a unique order ID and stores the order with status 'Placed'.	Creates the pickup request that pharmacy staff must process.
FR-004	Functional	The system shall notify the customer when the pharmacy admin marks the medicine order as ready for pickup.	High	A ready-for-pickup notification is sent/displayed within 30 seconds of the order status changing to 'Ready'.	Lets customers know when they can collect their medicines.
FR-005	Functional	The system shall allow password-protected pharmacy staff to update medicine inventory, add medicines, modify prices, and view daily sales reports.	High	An authenticated admin can perform each listed operation and the updated information is visible after saving.	Provides the staff functions specified in the scenario.
NFR-001	Nonfunctional	The system shall allow customers to place an order within 90 seconds.	High	At least 95% of valid test orders, measured from first interaction to successful submission, complete within 90 seconds.	Reduces waiting time at the medical store, especially during busy periods.
NFR-002	Nonfunctional	The system shall provide a simple and responsive customer interface.	Medium	At least 95% of interface actions respond within 2 seconds under normal operating load on supported devices.	Keeps the self-service experience easy to use and responsive.

2. Actors and Use Cases

Actors

• Customer

• Pharmacy Admin / Pharmacist

• Notification Service

Use Cases

ID	Use Case	Purpose
UC-01	Search & Select Medicines	Customer searches medicines and adds required items to the order.

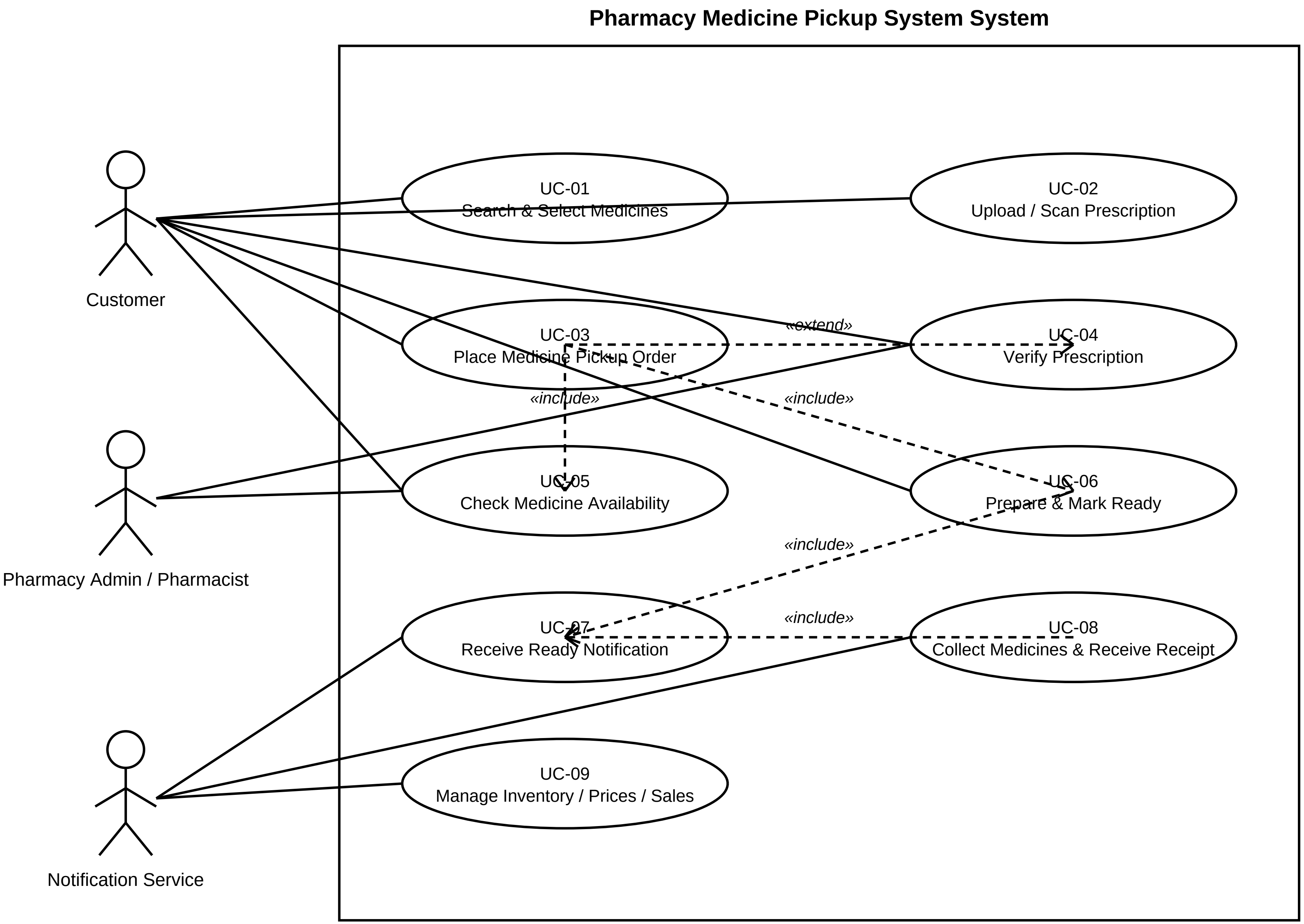
ID	Use Case	Purpose
UC-02	Upload / Scan Prescription	Customer provides a prescription when a medicine requires one.
UC-03	Place Medicine Pickup Order	Customer submits the selected medicines and required prescription.
UC-04	Verify Prescription	Pharmacist checks the prescription when applicable.
UC-05	Check Medicine Availability	Pharmacist verifies stock before preparing the order.
UC-06	Prepare & Mark Ready	Pharmacist prepares the order and marks it ready for pickup.
UC-07	Receive Ready Notification	Customer receives a notification when the order is ready.
UC-08	Collect Medicines & Receive Receipt	Customer collects the prepared medicines and receives a receipt.
UC-09	Manage Inventory / Prices / Sales	Admin manages medicines, prices, inventory, and daily sales reports.

3. UML Use-Case Diagram

The diagram shows the system boundary, actors, use cases, associations, and a conditional «extend» relationship.

Diagram is provided on the following page.

UML Use-Case Diagram – Pharmacy Medicine Pickup System



4. Use-Case Flow: Place Medicine Pickup Order

Main Success Scenario

1. Customer searches for the required medicine and selects the required quantity.

2. System displays the selected medicines in the order summary.

3. If any selected medicine requires a prescription, the system prompts the customer to upload or scan it.

4. Customer attaches the prescription, if required, and submits the pickup order.

5. System validates the order details and creates a unique order ID with status 'Placed'.

6. Pharmacy Admin verifies the prescription when applicable and checks medicine availability.

7. Pharmacy Admin prepares the medicines and marks the order as 'Ready for Pickup'.

8. System sends/displays a ready-for-pickup notification to the customer.

9. Customer collects the medicines and receives the receipt.

10. The use case ends successfully.

Alternate Flow

A1. Prescription required but invalid/missing: The system does not submit the order and asks the customer to provide a valid prescription.

A2. Medicine unavailable: The pharmacist marks the unavailable item; the system informs the customer and allows the order to be modified or cancelled.

A3. Notification failure: The order remains 'Ready for Pickup' and the system records the notification failure for staff follow-up.

Precondition: The system is operational and the customer can access the ordering interface.

Postcondition: A successful order is stored and the customer receives the appropriate confirmation; otherwise the system preserves the order state and provides an actionable error message.